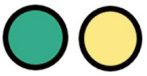
CHAPTER 20

Putting the TOGAF Standard into Practice



This chapter is part of the GREEN and GOLD reading tours. See Figure 1-1 in Chapter 1, Section 1.3, for alternative reading tours.

The TOGAF Standard describes in great detail the complete approach to an architecture implementation. All related aspects, such as adapting the Enterprise Metamodel and Content Framework, setting up the Architecture Repository, establishing architecture governance, and applying various techniques, are also richly described in the framework.

Applying all of this information in day-to-day practice still presents a challenge or two. To demonstrate that this need not be a showstopper, this chapter describes a case scenario in which all aspects of an architecture implementation are examined and executed using the Architecture Development Method. The case follows Jim, an Enterprise Architect for an organization called Academic Diagnostic Medicine.

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E. Jager, *Mastering the TOGAF® Standard*, https://doi.org/10.1007/979-8-8688-1814-1_20

20.1. A Case Scenario

Academic Diagnostic Medicine is a modern healthcare institution committed to delivering high-quality, patient-centered care while ensuring operational excellence and financial sustainability. In response to growing demands for value-based healthcare, the hospital has defined a strategic agenda for 2028–2030 that includes *improving patient satisfaction*, *optimizing internal workflows*, and *ensuring long-term financial health*.

To effectively realize this strategy, the hospital is engaging Jim, the Enterprise Architect it employs, to guide the transformation. Jim's role is to translate strategic drivers into executable capabilities, align business and IT domains, and ensure the change is planned and implemented in a structured, traceable, and value-driven manner.

Academic Diagnostic Medicine has been used to working with architecture for several years. As a result, the healthcare organization has a fairly high level of architecture maturity, as evidenced by the various strategic meetings Jim has had with the board. The TOGAF Standard has been the preferred standard since the introduction of working with architecture. Architecture work is done in accordance with the framework. Academic Diagnostic Medicine faces a number of challenges, and the board has decided that following and executing its newly defined strategy will allow them to realize their business goals.

20.1.1 Scoping the Work

March 2027–April 2027

During two strategy meetings with the board of directors, Jim learned about the plans that had been drawn up. Now, he has a clear understanding of what the upcoming period will entail. He visualizes the strategy using a Driver/Goal/Objective Diagram (Figure 20-1).

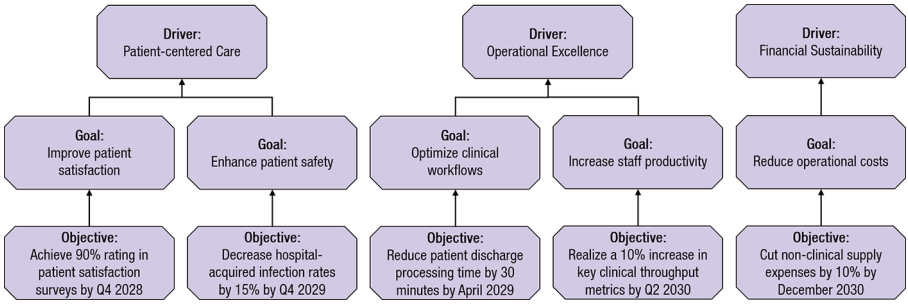


Figure 20-1. *Strategy of Academic Diagnostic Medicine*

The visualization suggests that implementing three major architectural projects can achieve the strategy. These projects appear to focus on the drivers and aim to improve patient satisfaction and safety, optimize clinical workflows and staff productivity, and reduce operational costs.

During a follow-up meeting with the board two weeks later, Jim explains the diagram he created earlier. He shares his vision for a phased, project-based approach, planning to divide the strategy into three projects.

The *improve patient satisfaction* project will focus on improving patient satisfaction and safety. The *improve clinical workflows* project will address improving workflow processes and will indirectly improve staff productivity. The third project will focus on *reducing operating costs*. The board recognizes the strategy and formally approves the diagram. The board also provides a set of requirements for the work. Jim records these in a Requirements Catalog (see Table 20-1).

Table 20-1. *Requirements Catalog*

Project	Requirement	Goal
Improve patient satisfaction	A centralized digital feedback platform needs to be made available that enables real-time collection and routing of patient satisfaction and safety-related feedback to appropriate departments.	Improve patient satisfaction. Enhance patient safety.
Optimize clinical workflows	Clinical workflows must be digitized and integrated into the hospital's EHR system, allowing automated role-based task assignments.	Optimize clinical workflows. Increase staff productivity.
Reduce operational costs	A cost-tracking dashboard must be implemented that links departmental spending to patient outcomes, enabling quarterly reviews of cost-effectiveness.	Reduce operational costs.

Jim directly links the requirements to the goals of Academic Diagnostic Medicine. This makes the requirements for achieving the goals clear to everyone at all times.

During the implementation of the strategy, Jim anticipates that a number of risks will arise that need to be identified. Jim is therefore considering adding the risk entity to the Enterprise Metamodel used by Academic Diagnostic Medicine. The inclusion of this new entity will allow Jim to define and capture risks in a Risk Catalog. From this artifact, they can be visualized as needed.

20.1.2 Envisioning the Architecture

May 2027–June 2027

The board's approval of the strategy visualization is the green light for Jim to begin his first inventory. He starts talking to different people in the

organization. The reason for this is to get as complete a picture as possible of the work to be done. To keep track, Jim creates an initial Stakeholder Catalog (see Table 20-2). This artifact initially contains the names, roles, and classifications of the stakeholders with whom he will interact.

Table 20-2. *Stakeholder Catalog*

Stakeholder	Classification
Maria Summers (CEO)	Keep satisfied
Chris West (CIO)	Key player
James Noble (CFO)	Keep satisfied
Anna Carter (Manager HR)	Minimal effort
Susan Delgado (CNO[1])	Key player
Benjamin Lee (CMIO[2])	Key player
Elena Petrova (Medical Director of Quality and Safety)	Key player
Kevin O'Brien (Patient Experience Manager)	Key player
Hannah Duarte (Charge Nurse/Unit Manager)	Keep informed

[1] Chief Nursing Officer.

[2] Chief Medical Information Officer.

Jim talks to everyone directly involved. In addition to the board of directors, this includes various executives, clinical staff, and various operational teams. Jim wants to understand how the strategy is perceived at the operational level. Gradually, a picture emerges of the best way to communicate with the various stakeholders. Every time Jim finds a way that works well, he records it in the Communications Plan. He can use this plan throughout the execution of the work to communicate with all stakeholders. The different conversations reveal the stakeholders' concerns and constraints (see Table 20-3).

Table 20-3. *Strategy concerns and constraints*

Goal	Concern	Constraint
Improve patient satisfaction		Low digital literacy for older patients
Enhance patient safety	Will feedback loops criticize care rather than improve it?	
Optimize clinical workflows	Will workflow changes require system re-training?	
Increase staff productivity		Resistance to process standardization
Reduce operational costs	How fast will cost savings materialize?	

The concerns and constraints are all things he needs to consider. Jim adds the concerns to the Stakeholder Catalog that he created earlier.

During one of Jim's weekly board meetings, he provides feedback on the status of the process. He indicates that he now has an understanding of the stakeholders involved and their concerns and constraints. All of this information, along with the requirements established by the

board, is captured in a Statement of Architecture Work. This document contains a high-level overview of the work, in addition to the components mentioned above.

20.1.3 Starting the Development

July 2027

To get a good idea of what exactly needs to be done, Jim turns to the existing content of the Architecture Repository. This helps him determine what the current situation looks like. A quick look at the documented material from previous projects provides building blocks in the form of a Business Capability Map (Table 20-4) and a Capability/Business Process Matrix (Table 20-5) that can be reused.

Table 20-4. *Business Capability Map*

Capability (Level 1)	Capability (Level 2)	Description
Patient Engagement	Patient Feedback Management	Ability to collect, analyze, and respond to feedback
Clinical Services	Clinical Workflow Optimization	Ability to manage patient care workflows efficiently
	Workflow Coordination	Ability to coordinate patient care workflows
Operations	Resource Scheduling	Ability to align staff and resource schedules
Financial Management	Cost Reduction Management	Ability to manage cost effectively

The Capability/Business Process Matrix is a tailored artifact that he created in the past. It contains an overview of the existing processes in the organization that are related to the capabilities.

Table 20-5. *Capability/Business Process Matrix*

Capability (Level 2)	Business Process
Patient Feedback Management	Feedback Collection
Clinical Workflow Optimization	Optimized Discharge Planning
	Surgical Workflow Coordination
Resource Scheduling	Staff Roster Optimization
	Equipment Usage Scheduling
Cost Reduction Management	Procurement

Jim uses the space provided for this task to get a good sense of Academic Diagnostic Medicine's strategy. He creates a Strategy/Capability Diagram (see Figure 20-2) by combining the Driver/Goal/Objective Diagram with the Business Capability Map. This new artifact ensures that he has a talking point when discussing the impact of the strategy within the organization.

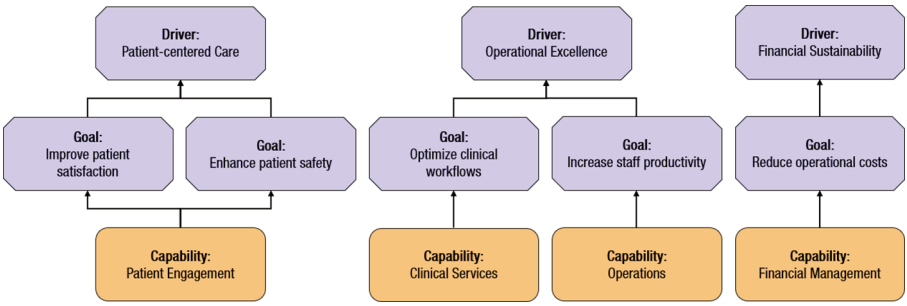


Figure 20-2. *Strategy/Capability Diagram*

The relationship between strategy and capabilities makes it easier for many stakeholders to understand where the strategy impacts the organization and what it means for them. This allows Jim to show at a glance where the challenges lie in the context of the strategy.

All the information gathered so far is added to the Statement of Architecture Work document. Jim has this document formally approved by the board and then shares it with the relevant stakeholders. They also sign off on the document.

August 2027–October 2027

With all the available and retrieved information in hand, Jim gets to work. His first goal is to establish the Baseline Business Architecture for each of the identified capabilities. The baseline consists of a number of pre-existing building blocks. These include capabilities, business processes, and the associated organizational structure. By iteratively going through the architecture process, Jim is able to establish the Baseline Architecture for each of the identified capabilities. One of the tools he uses is a Business Capability Map. Using this artifact, he will be able to easily identify where pain points (such as missing capabilities) exist in the current situation. The missing capabilities are shown in the Business Capability Map in Table 20-6.

Table 20-6. *Business Capability Map listing missing capabilities*

Capability (Level 1)	Missing Capability (Level 2)	Description
Patient Engagement	Feedback Analysis	Ability to analyze patient satisfaction feedback
Financial Management	Supply Chain Management	Ability to manage and rationalize the supply chain

The lack of Level 2 capabilities for Patient Engagement and Financial Management indicates that there are no processes in place to analyze feedback and streamline the supply chain. The missing processes are shown in Table 20-7.

Table 20-7. *Capability/Business Process Matrix listing missing processes*

Capability (Level 2)	Missing Process
Patient Feedback Management	Feedback trend analysis
	Infection prevention audit
Cost Reduction Management	Supply chain rationalization

Jim's mapping of existing roles in the organization, using an Organization/Business Roles Catalog (see Table 20-8), shows that there are currently no appropriate roles to perform the new processes. Jim makes a note to himself to bring this issue to the attention of HR. Perhaps one or more vacancies will need to be advertised to fill the current staffing gap.

Table 20-8. *Organization/Business Roles Catalog*

Organization Unit	Business Role	Status of Role
Board of Directors	Chief Executive Officer	Existing
Finance	Chief Financial Officer	Existing
	Financial Analyst	Non-existent
Operations	Chief Operations Officer	Existing
IT Department	Chief Information Officer	Existing
Quality and Safety	Medical Director	Existing
	Patient Safety Officer	Non-existent
	Patient Experience Manager	Existing
Clinical Operations	Chief Nursing Officer	Existing
	Chief Medical Information Officer	Existing
	Scheduling Coordinator	Existing
	Charge Nurse/Unit Manager	Existing

Once the Baseline Business Architectures have been established, Jim can use the established strategy to determine what the Target Architectures should look like. Again, the building blocks from the Architecture Repository come in handy. Jim recognizes the importance of aligning the newly created business capabilities with the goals of Academic Diagnostic Medicine. To convince the board of the need, he uses a Capability Heat Map (Figure 20-3), which clearly shows which capabilities are working properly and which are not. The artifact also shows where new capabilities are needed.

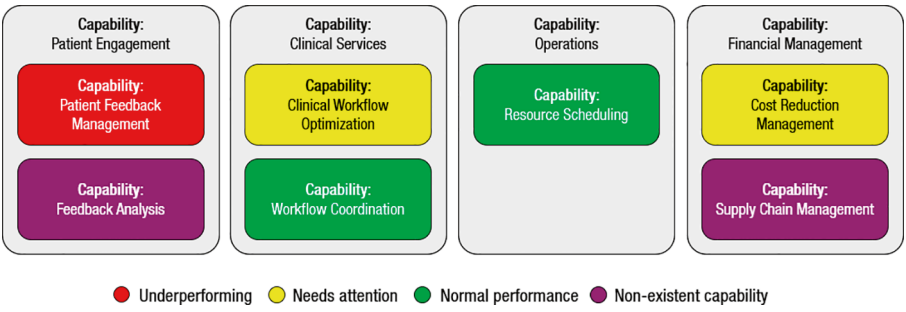


Figure 20-3. *Capability Heat Map*

The Patient Feedback Management capability, in particular, requires attention. It is underperforming and must be addressed. The Clinical Workflow Management and Cost Reduction Management capabilities will also need to be examined more closely and improved where possible. This can be achieved by automating the processes associated with these capabilities. Currently, the Clinical Workflow Management capability uses manually performed tasks by a specific business role. The other capability seems to focus exclusively on order processing. Much less attention is paid to mapping the entire supply chain. To address this, two new capabilities, Feedback Analysis and Supply Chain Management, will have to be created. These new capabilities will support Patient Engagement and Financial Management, respectively.

After mapping the capabilities, Jim focuses on the Baseline and Target Architectures for the Application and Technology segments. He uses a Process/Application Realization Diagram (see Figure 20-4) to illustrate the current connection between the application landscape and business processes.

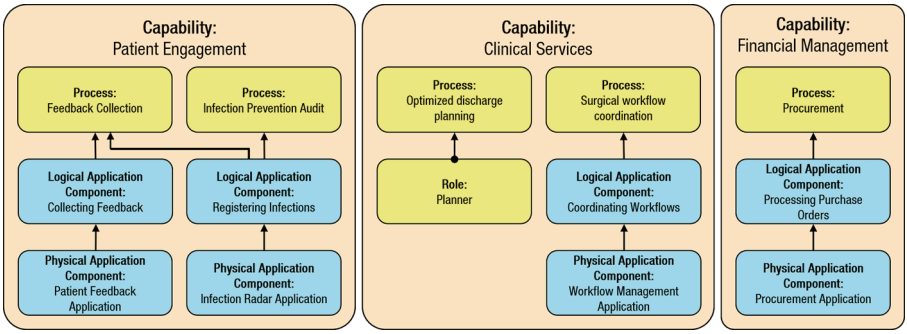


Figure 20-4. *Process/Application Realization Diagram showing Baseline Application Architecture*

As the diagram shows, there is currently no functionality to analyze incoming feedback. Incoming feedback is simply stored in a system. No analysis is performed on it. The board of directors has indicated the need for a central digital feedback platform that can process and route patient satisfaction and safety-related feedback in real time to the relevant departments.

Within the Clinical Services capability, discharge planning is currently managed manually by the Planner business role. One of the board's requirements indicates that this process should be automated.

For the Financial Management capability, the current method of processing purchase orders is inadequate. A cost-tracking dashboard must be made available to provide insight into the costs incurred by each department. To this end, the supply chain will also need to be examined.

Based on the stated requirements, Jim creates a visualization of the possible Target Application Architecture (see Figure 20-5). He fleshes out the requirements by creating an application function that analyzes incoming feedback. Jim also replaces the manual process within the Clinical Services capability with an automated variant that provides real-time mapping of departmental expenditures. The Supply Chain Management Application and the Managing Supply Chain application function address this.

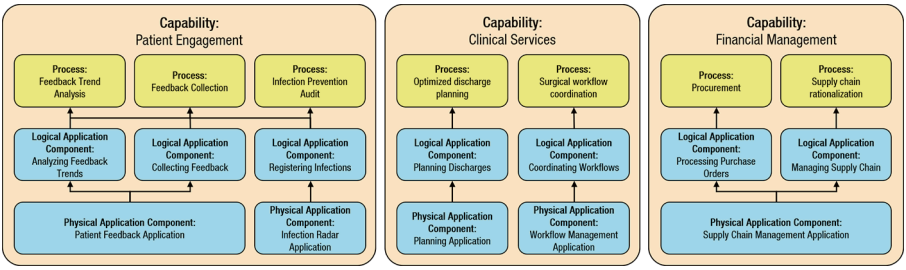


Figure 20-5. *Process/Application Realization Diagram showing Target Application Architecture*

Jim was recently informed by the COO that no changes will be made to the existing technical infrastructure. Academic Diagnostic Medicine invested heavily in a cloud-based setup last year and migrated all its services to it. There is no budget to make adjustments. Therefore, Jim will have to adapt the architecture to the infrastructure and services currently available.

Jim draws up an initial version of the Architecture Definition Document. In it, the main objectives and scope of the architecture are clearly defined. He makes sure the document contains detailed descriptions of the Baseline and Target Architectures.

20.1.4 Setting the Direction

November 2027

Of course, the transition from the Baseline Architecture to the Target Architecture will not happen all at once. Jim assumes that two Transition Architectures will be needed as intermediate phases. He creates a Plateau Planning Diagram, shown in Figure 20-6, in which the Baseline, Transition, and Target Architectures are visible. He also uses the diagram to visualize the gaps between the architectures. These gaps will need to be closed by carrying out a number of initiatives.

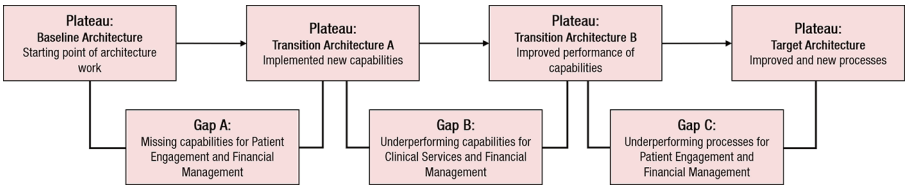


Figure 20-6. *Plateau Planning Diagram*

Closing the gaps requires multiple work packages. Work packages consolidate well into projects, which is an advantage. With the help of a project leader, the required activities can be fleshed out. To explain these activities to the project leader, Jim compiles a Work Package Portfolio Map (see Table 20-9). In this artifact, he lists all the work packages, the gaps they close, the associated initiatives (activities), and the goals to which the work packages are linked.

Table 20-9. *Work Package Portfolio Map*

Work Package Name	Gap	Activity	Associated with Goal(s)
Implement new capability for Patient Engagement	A	Implement new capability Feedback Analysis.	Improve patient satisfaction.
Implement new capability for Financial Management	A	Implement new capability Supply Chain Management.	Reduce operational costs.
Improve capability for Clinical Services	B	Improve capability Clinical Workflow Optimization.	Optimize clinical workflows.
Improve capability for Financial Management	B	Improve capability Cost Reduction Management.	Reduce operational costs.
Implement and improve processes for Patient Engagement	C	Implement process Feedback trend analysis.	Improve patient satisfaction.
	C	Improve process Feedback collection.	
	C	Implement process Infection prevention audit.	
Improve processes for Financial Management	C	Improve process Supply chain rationalization.	Reduce operational costs.

Jim begins to build a first draft of the Architecture Roadmap based on the gaps and corresponding initiatives captured in work packages. These work packages form the basis of the draft roadmap and are a welcome addition. Jim understands that the roadmap does not need to be finalized at this stage of the architecture development process. It will evolve as the architecture work progresses. New business capabilities and processes are added to the roadmap in the form of work packages. Jim pays close attention to aligning these components with organizational

goals and objectives. Jim takes extra care to map the work packages to the timeframes mentioned in the corresponding objectives.

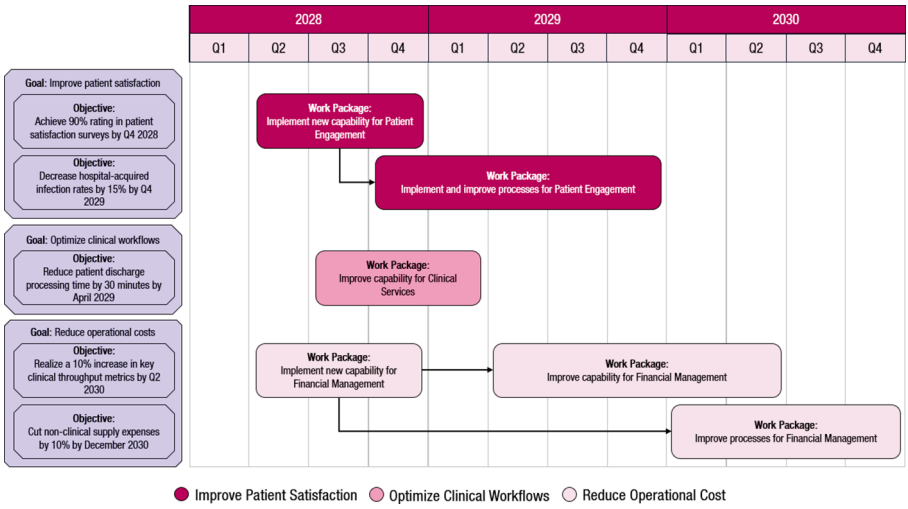


Figure 20-7. *Draft Architecture Roadmap containing work packages*

Figure 20-7 illustrates the relationship between the work packages and the goals and objectives. The colors indicate which work packages belong to which project.

Armed with the roadmap, Jim visits the HR department. He discusses how the absence of certain roles complicates the realization of the strategy. These roles include the newly created positions of Patient Safety Officer in the Quality and Safety Department and Financial Analyst in the Finance Department. After consulting with an HR business partner, Jim determines that filling these roles is necessary. The HR business partner takes the initiative to contact the board of directors and coordinate posting the needed vacancies. Jim will be informed of the outcome.

December 2027

One of Jim's next steps is to determine the impact of the architecture work on the existing one. He meets with the project manager, who is already aware of the impending work. After a brief discussion, they conclude that there is currently no reason to believe the new architecture will interfere with ongoing projects, or vice versa.

Over the next week, Jim records all the information he has gathered into a rough draft of the Architecture Definition Document. He also uses this time to schedule appointments with various stakeholders. He wants to explain the upcoming work so that everyone involved has a clear understanding of what is to come.

During the group sessions, Jim discusses the current state of affairs and the direction the organization has chosen. He explains that the recorded Target Architecture will be realized using Plateau Planning, a phased approach. Several business stakeholders, executives, process owners, and subject matter experts are present during the work sessions. Their insights are invaluable in shaping an architecture that reflects the organization's needs and goals.

Jim reviews the requirements provided by the board in front of the stakeholders. After the session, all stakeholders agree that these should be the criteria for the architectural work.

Jim determines the necessary building blocks, or capabilities, based on the information available to him. Then, he performs a final cross-check to ensure that the architecture aligns with the organization's overarching business goals. This alignment is vital for the architecture's effectiveness and relevance. The reasoning behind every decision made during this process is documented. This practice maintains a clear link between architectural choices and organizational objectives.

The Baseline, Transition, and Target Architectures are described in detail and added to the Architecture Definition Document. The same goes for the gaps between the architecture states. The Driver/

Goal/Objective Diagrams, Business Capability Map, and Capability/Business Process Matrix are also included in the Architecture Definition Document. The document also includes the Plateau Planning Diagram and the Process/Application Realization Diagrams for the Baseline and Target Architectures. A draft version of the Architecture Roadmap is also incorporated. These diagrams, map, and matrix effectively communicate key aspects of the architecture. Visual representations simplify complex information, making the document more accessible to stakeholders.

After finishing the first draft of the document, Jim shares the Architecture Definition Document with key stakeholders to gather their insights and feedback. He then uses this feedback to make the necessary adjustments to the document. After two short iterations, everyone agrees that the written plan is what needs to happen. Any concerns raised by stakeholders have been addressed, and everyone feels that this approach will align with organizational objectives.

20.1.5 Getting Ready for Implementation

January 2028–February 2028

The next step in the architecture process is creating an Implementation and Migration Plan. To do so, Jim must first identify any existing factors that could impact the execution of the architectural work. Only three factors that will potentially affect the work emerged during the preparation of the Implementation Factor Catalog (see Table 20-10).

Table 20-10. *Implementation Factor Catalog*

Factor Category	Factor	Description	Deduction
Effect (impact)	Change in business process	Automation of discharge planning	New application will automate work done by hand.
Action	Introduction of new Planning Application	Implementation of new application	Planner role needs to be trained to use the new application.
Assumption	Introduction of Feedback Analysis application function	Implementation of new application functionality	Feedback analysis • may result in both negative and positive feedback • needs to be able to distinguish between both types of feedback

Implementing the new Planning Application will replace the Planner role, which is currently responsible for manually executing the process, with an automated variant. This means that those filling the role will need to be trained to use the new application. Additionally, the new application's functionality will provide greater visibility into feedback received from patients. This may include criticism, in addition to positive feedback. The Patient Feedback Application must be able to distinguish between these two types of feedback. Jim realizes that the three risks that emerged from the deduction of the Implementation Factor Catalog may affect the implementation of the Target Architecture. He makes a note to himself to properly identify and clearly visualize the risks. There is an opportunity to do this when determining the business value of the projects

to be implemented. Fortunately, he included risk as an entity in the Enterprise Metamodel at an earlier stage.

To adequately flesh out the work to be performed and establish a sound architecture, Jim reviews the earlier noted concerns and constraints.

One of the identified constraints is the low digital literacy of older patients. Fortunately, little to nothing is changing for patients at this time. The changes are more on the back end of the process. The Target Architecture will change how incoming feedback is processed and analyzed. This will address the aforementioned constraint.

A second constraint is the possible resistance to process standardization. This mainly concerns optimizing the clinical workflow process. Since the current manual process will be automated and a new application will be introduced for this purpose, it is crucial to provide thorough training to users of the new application. Once users feel confident using the application, their resistance to standardizing the process will diminish, if not disappear.

The risk analysis should address the concern that the feedback process will generate more criticism than valuable information. Although there is little change on the patient side, negative feedback is still a possibility. If the analyses show this, Academic Diagnostic Medicine will have to take action.

The concern about whether staff training is needed is also valid. However, this can easily be overcome by providing training for the Planner role, which will work with the newly implemented application. Other than that, no customization is required.

Finally, the concern about how quickly cost savings will be visible cannot be addressed at this time. In any case, cost savings are not the primary objective. First, the board of directors wants insight into costs per department. Cost reduction is not planned until the end of 2030.

After ensuring that the concerns and constraints have been adequately addressed, Jim prepares an initial draft of the Implementation and Migration Plan, which is presented in Table 20-11. He divides the plan into

three phases corresponding to the two Transition Architectures and the Target Architecture.

Table 20-11. *Implementation and Migration Plan*

Phase	Timeframe	Work Packages	Key Milestones	Dependencies
1	Q2 2028 to Q4 2028	Implement new capability for Patient Engagement.	Implemented new capability	–
		Implement new capability for Financial Management.	Implemented new capability	–
2	Q3 2028 to Q2 2030	Improve capability for Clinical Services.	Improved performance of capability	Staff training complete
		Improve capability for Financial Management.	Improved performance of capability	–
3	Q4 2028 to Q4 2030	Implement and improve processes for Patient Engagement.	Improved and new process	Implemented new capability for Patient Engagement
		Improve processes for Financial Management.	Improved process	Improved capability for Financial Management

Jim uses the Plateau Planning Diagram (Figure 20-6) to define the phases. The Work Package Portfolio Map (Table 20-9), which he created earlier, comes in handy. In it, the work packages are neatly grouped and described, including the gaps that will be closed by executing the activities. The goals that the work packages will accomplish are also listed.

Since the work packages and their associated activities relate to the gaps, it is easy to identify the work packages' dependencies. For example, Jim emphasizes the importance of carefully implementing the two new capabilities for Patient Engagement and Financial Management (Feedback Analysis and Supply Chain Management, respectively). These new capabilities are prerequisites for later steps. In the Implementation and Migration Plan, Jim takes this into account.

During an earlier meeting with the project leader, it was revealed that there are currently no ongoing projects that would prevent the implementation of the architecture. Nevertheless, to avoid any surprises, Jim contacts the project leader again. During their conversation, they confirm again that there are no overlapping projects that could limit the implementation of the new architecture.

Now that everything seems to be in order, Jim conducts a Business Transformation Readiness Assessment (see Table 20-12). This assessment helps Jim determine if there are any roadblocks to the organization's readiness for implementation.

Table 20-12. *Business Transformation Readiness Assessment*

Readiness Factor	Assessment Criteria	Score (1–5)
Leadership commitment	Level of commitment from top leadership	5
Employee engagement	Willingness and enthusiasm of employees for change	3
Organizational culture	Openness to new ideas and adaptability	5
Skills and competencies	Availability of required skills and competencies for the transformation	3
Risk Management	Identification and mitigation of potential risks	4
Stakeholder alignment	Alignment of stakeholders' interests and expectations	5

The assessment reveals that both employee engagement and the necessary skills and competencies require attention. This is likely related to the automation of tasks that are currently performed manually. Specifically, the *optimized discharge planning* process within the Clinical Services capability will be automated. After the implementation of the architecture, specialized skills will be required to run these tasks through the automated process. Training the relevant personnel is one possible solution.

Jim schedules another meeting with the relevant stakeholders and presents the migration plan. A few questions are asked about the order of the planned activities, which Jim can easily answer using the Plateau Planning Diagram and the Work Package Portfolio Map to explain how the order was established. The stakeholders unanimously approve the Implementation and Migration Plan.

Later, Jim runs into the HR business partner he spoke with a few weeks ago. She was just on her way to see him with good news. The positions for Patient Safety Officer and Financial Analyst have been filled.

March 2028–May 2028

Jim asks the project manager overseeing the execution of the work to incorporate relevant parts of the Implementation and Migration Plan into the appropriate project management framework. Jim also creates a Business Value and Risk Matrix to illustrate the business value of the projects to be implemented (see Figure 20-8).

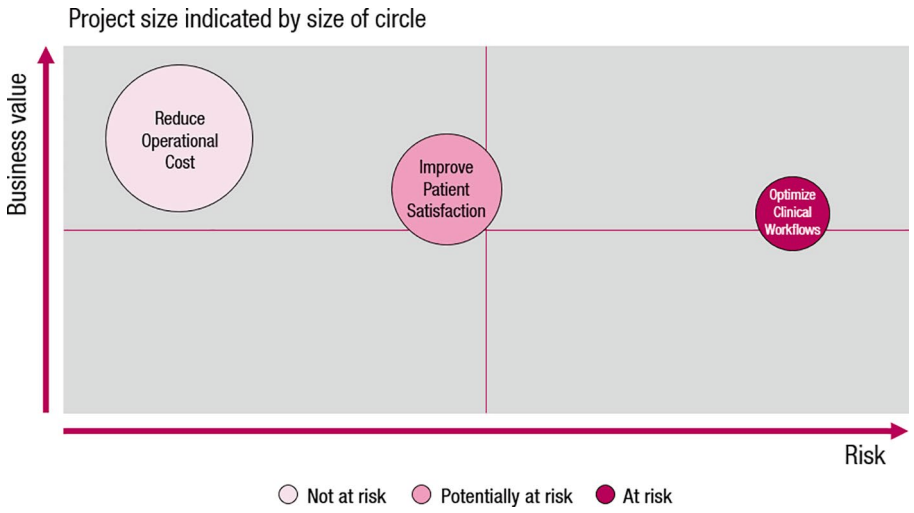


Figure 20-8. *Business Value and Risk Matrix*

The matrix shows the three previously defined projects: reduce operational costs, improve patient satisfaction, and optimize clinical workflows. Jim captures the risks he noticed earlier when creating the Implementation Factor Catalog in a Risk Catalog. Then, he plots the risks on the projects. The project with the highest risk is the *optimize clinical workflows* project. This is because it involves automating tasks that are currently performed manually. Those responsible for executing these tasks will need to reinterpret the Planner role. They will also need to undergo training to use the Planning Application correctly. The *improve patient satisfaction* project has a significantly lower risk because the only question is whether making feedback more available will lead to more criticism.

Jim combines the previously created Risk Catalog with the Process/Application Diagram and uses a Risk/Process/Application Realization Diagram to visualize the risks, as shown in Figure 20-9.

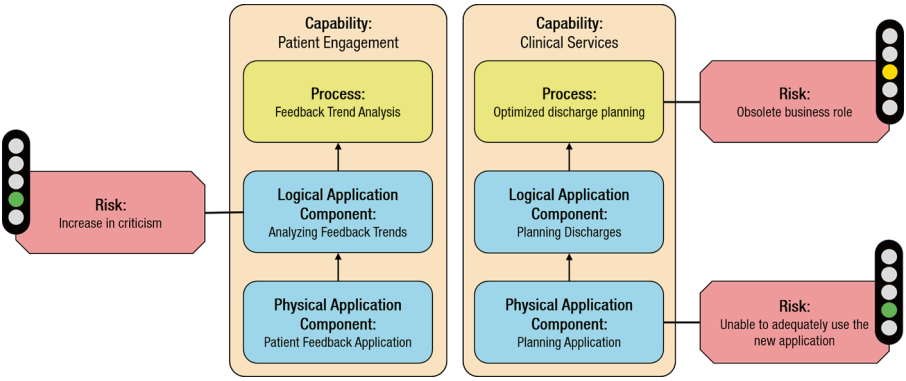


Figure 20-9. *Risk/Process/Application Realization Diagram*

The figure shows the two capabilities for which one or more risks have been identified. These risks have been assigned to the corresponding building blocks of the planned architecture. Each risk has also been assigned a score, which has been visualized using a traffic light. The five color levels in the graphic represent the five risk levels used in the Risk Management technique. Jim draws up a set of risk mitigation measures and submits them to the relevant stakeholders. Along with the latest versions of the Architecture Roadmap and the Implementation and Migration Plan, Jim obtains formal approval for the work to be performed.

20.1.6 Overseeing the Architecture Work

June 2028–February 2029

The work is then carried out according to the terms of the Architecture Contract. Jim supervises the execution of activities and the implementation of the overall architecture. He regularly performs random checks to ensure that the agreements are still being met. He shares the Architecture Compliance Report with the involved stakeholders so they are well-informed.

March 2029–August 2029

Some time later, Jim receives confirmation that the new capabilities, part of Transition Architecture A, have been implemented and improved. After briefly checking the delivery, Jim concludes that everything went according to plan. He updates the Business Capability Map and presents an updated Capability Heat Map (see Figure 20-10).

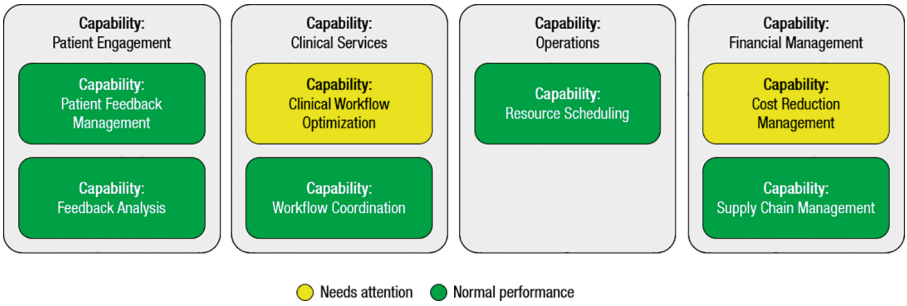


Figure 20-10. *Updated Capability Heat Map*

The Capability Heat Map shows the addition of Feedback Analysis and Supply Chain Management capabilities. Both capabilities are operating satisfactorily. Furthermore, implementing the former capability has positively impacted Patient Feedback Management.

20.1.7 Maintaining Control

September 2029–July 2030

Jim regularly meets with the most involved stakeholders and asks them about the state of affairs using questionnaires. This gives him a good picture of how the architecture is being implemented. This also enables him to make adjustments where necessary. Based on the performance reviews he conducts, Jim effectively manages the architecture change management process.

Around January 2030, Jim receives a message from the board of directors. They inquire about the effectiveness of his work and ask him to

explain it in person. To prepare for the upcoming consultation, Jim uses the Driver/Goal/Objective Diagram with the heat mapping technique (see Figure 20-11). This technique allows him to easily visualize where progress has been made and where attention is still needed in the coming period.

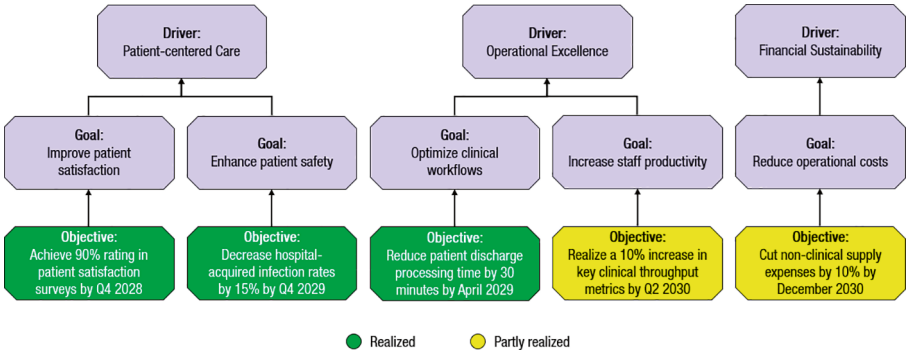


Figure 20-11. *Driver/Goal/Objective Diagram Heat Map*

The heat map of the Driver/Goal/Objective Diagram shows that not all objectives have been achieved yet. In particular, increasing key clinical throughput and reducing non-clinical supply expenses have not yet been fully realized. This is not surprising, since the year 2030 has not yet ended. Nevertheless, Jim is conducting an investigation to determine if any unforeseen issues have arisen.

The investigation quickly revealed that the learning curve for the new Planning Application used in Clinical Operations was steeper than initially assumed. This delayed the adoption of the new way of working. Consequently, the goal of a 10% increase in throughput has not yet been achieved. However, it is expected to be achieved by the end of Q2 2030.

Regarding the reduction of operating costs, the survey shows that the targeted 10% reduction has not yet been achieved. Initially, however, the board focused on identifying where expenses were occurring. As a result, more attention was given to recording expenditures than to reducing costs. Jim concludes that this objective will be achieved in a timely manner.

August 2030

Several months pass once again, and the architecture is approaching its final form. Significant efforts are currently being made to enhance the processes employed in capabilities. This will be the final phase that achieves the Target Architecture.

Jim invested a lot of time preparing for the work and ensuring that Academic Diagnostic Medicine would succeed in realizing its strategy. Thanks to this successful implementation, Enterprise Architecture's ability to add value to an organization has once again been demonstrated.

20.2. Summary

In the case scenario described in this chapter, Enterprise Architect Jim demonstrated how the TOGAF Standard can be used to develop and implement an architecture through a tailored Architecture Development Method.

He covered all phases of the Architecture Development Method while developing and implementing the architecture. Some phases were iterative and thus performed repeatedly. At other times, Jim briefly transitioned to subsequent phases, only to return almost immediately to the current phase. The TOGAF Standard's adaptability allows for this flexibility, making it possible to develop and implement architecture in a way that suits different scenarios, organizations, and industries.

March 2027–April 2027 (Scoping the Work)

Preliminary Phase

Table 20-13. *Steps performed during the Preliminary Phase*

Steps Performed	Description of Step
Identify key drivers and organizational context	Pinpointing the critical factors that influence the architecture, such as key drivers and the organizational context.
Assess Enterprise Architecture maturity	Evaluating where the organization stands in its architectural development.
Choose and tailor the architecture framework	Identifying and tailoring the TOGAF Standard, Architecture Development Method, and the Content Framework.

During the Preliminary Phase (see Table 20-13), Jim used the Driver/Goal/Objective Diagram to visualize the key drivers and organizational context. He reviewed the diagram with the most important stakeholders at the time, the board of directors, and coordinated with them.

Since Academic Diagnostic Medicine has been working according to the TOGAF Standard for years, Jim saw no need to conduct an architecture maturity measurement. The TOGAF Standard has been considered the preferred framework for working with architecture since its introduction. Later in the project, Jim adapted the framework by adding the risk entity to the Enterprise Metamodel. He also incorporated the Risk Catalog as an artifact in the Content Framework.

May 2027–June 2027 (Envisioning the Architecture)
Phase A: Architecture Vision

Table 20-14. *Steps performed during Phase A*

Steps Performed	Description of Step
Establish the architecture project	Obtain formal project recognition, secure endorsement from corporate leadership, and gain support and commitment from relevant line management.
Identify stakeholders and determine requirements	Identify stakeholders and requirements to allow for the architecture and related work to be communicated and aligned.
Create a high-level architecture vision	Develop a high-level description of the architecture solution that will meet business requirements and outline how it aligns with the overall business strategy.
Develop Statement of Architecture Work	Develop Statement of Architecture Work and secure formal approval.

During Phase A (see Table 20-14), Jim defined and outlined the projects to be implemented. Three project initiatives emerged based on the strategy. Jim sought and received formal board approval for the project format.

He identified the key stakeholders he would encounter during the development and implementation of the projects. In doing so, he considered the requirements established by the board of directors and the concerns and constraints raised by the stakeholders.

Based on all the input he collected, Jim determined which capabilities would be the subject of the work and visualized this using a Strategy/Capability Diagram. This diagram shows the high-level architecture vision. Jim documented the high-level work to be performed in a Statement of Architecture Work. This document includes stakeholder concerns and constraints, as well as the requirements set by the board of directors. The board of directors and the stakeholders formally approved the document.

July 2027–October 2027 (Starting the Development)
Phase B: Business Architecture
Phase C: Information Systems Architectures
Phase D: Technology Architecture

Table 20-15. *Steps performed during Phases B, C, D*

Steps Performed	Description of Step
Select reference models, viewpoints, and tools	Choose relevant models and viewpoints from the Architecture Repository to match the business goals and stakeholder interests.
Develop Baseline and Target Business Architecture description	Describe the current and future Business Architecture to a degree that supports the architecture vision, building on the Architecture Repository's resources as needed.
Perform gap analysis	Confirm internal consistency, identify any gaps between baseline and target, and validate that models align with business objectives.
Define candidate roadmap components	Using the baseline, target, and gap analysis, build an initial roadmap to guide future actions and prioritization across the remaining architecture phases.
Conduct formal stakeholder review	Revisit the architecture's initial motivation and make any necessary refinements to ensure it's fit to support other architecture domains.
Create the Architecture Definition Document	Complete all relevant sections in the Architecture Definition Document, adding models, reports, and graphics as needed to capture a clear architectural view.

Jim used existing material from the Architecture Repository. He extracted the previously defined building blocks, the Business Capability Map and the Capability/Business Process Matrix, for reuse.

Using these building blocks, he developed the Baseline and Target Architectures. Jim performed a gap analysis to define the candidate building blocks for the Architecture Roadmap.

Throughout the process, Jim coordinated with the stakeholders involved through formal stakeholder reviews. After establishing the Baseline and Target Architectures, Jim prepared an Architecture Definition Document. In the document, he outlined the detailed designs of the Baseline and Target Architectures (see Table 20-15).

November 2027–December 2027 (Setting the Direction)
Phase E: Opportunities and Solutions

Table 20-16. *Steps performed during Phase E*

Steps Performed	Description of Step
Determine/confirm key corporate change attributes	Align the architecture with an organization's business culture. This involves creating an Implementation Factor Catalog and assessing the transition capabilities of organizational units.
Consolidate and reconcile interoperability requirements	Identify and address constraints that may hinder interoperability.
Identify and group major work packages	Assess and address gaps in business capabilities in order to enable the organization to effectively plan and implement necessary changes aligned with strategic objectives.
Identify Transition Architecture(s)	Identify Transition Architecture(s) to provide structured, value-driven steps toward achieving the Target Architecture, tailored to the organization's implementation strategy and capacity for change.
Create Architecture Roadmap and Implementation and Migration Plan	Create a structured approach to transition from Baseline to Target Architectures through a detailed Architecture Roadmap and Implementation and Migration Plan.

Jim used an Implementation Factor Catalog to identify potential risks in the architecture work (see Table 20-16). He also used the catalog to identify areas of the organization with limited change capacity. The architecture to be implemented has been divided into several phases. Each phase represents a Transition Architecture. Jim indicated which activities (initiatives) were associated with each Transition Architecture. These activities were consolidated into work packages and used as plannable components in the Architecture Roadmap. Based on the roadmap, Jim drew up an Implementation and Migration Plan.

January 2028–May 2028 (Getting Ready for Implementation)
Phase F: Migration Planning

Table 20-17. *Steps performed during Phase F*

Steps Performed	Description of Step
Confirm management framework interactions	Align the Implementation and Migration Plan with an organization's management frameworks.
Assign business value to work packages	Define and measure business value for work packages, identify corresponding projects, assess associated risks, and estimate business value.
Prioritize migration projects through cost/benefit assessment and risk validation	Determine the sequence of projects based on business priorities, technical dependencies, resource availability, and risk factors.
Confirm Architecture Roadmap	Update the Architecture Roadmap, incorporating any necessary Transition Architectures.
Complete Implementation and Migration Plan	Develop a detailed timeline with clear milestones for each transition stage.

Since Jim wanted a project manager to coordinate the work, aligning the architectural approach with the project management framework was useful. The TOGAF Standard can easily be linked to and aligned with other frameworks.

Jim assigned a business value to each work package. He used the Business Value and Risk Matrix he created earlier to do so. He also managed to visualize the risks using a Risk/Process/Application Realization Diagram. The architecture developed thus far has been checked and verified with the stakeholders (see Table 20-17).

June 2028–August 2029 (Overseeing the Architecture Work)
Phase G: Implementation Governance

Table 20-18. *Steps performed during Phase G*

Steps Performed	Description of Step
Confirm scope and priorities	Identify and address gaps within the existing architecture. Review the Work Package Portfolio Map to determine which initiatives lead to optimal value extraction.
Perform architecture compliance reviews	Regularly assess the implementation activities to ensure that they are consistent with the architectural standards and principles.
Perform post-implementation review	Examine the end result of the architecture work. If there are deviations, submit change requests to correct them.

Jim took on a leading role during the execution of the work. As an Enterprise Architect, he did not need to operate the controls himself. From his coordinating position, he oversaw the implementation of the work.

The Work Package Portfolio Map, which had been drawn up previously, provided Jim with the tools necessary to prioritize the activities (initiatives), ensuring that the work was carried out in the correct order. Throughout the implementation process, Jim conducted random

compliance reviews to ensure everything was on track. He reviewed the results with the relevant stakeholders repeatedly.

After each successful implementation of a Transition Architecture, he conducted a post-implementation review (see Table 20-18).

September 2029–July 2030 (Maintaining Control)
Phase H: Architecture Change Management

Table 20-19. *Steps performed during Phase H*

Steps Performed	Description of Step
Deploy monitoring tools	Deploy tools to monitor the business and technology changes that could impact the architecture, as well as the maturity of the Architecture Capability.
Manage risks	Identify, classify, and mitigate risk(s) before business transformation.
Provide analysis for architecture change management	Analyze information from monitoring tools, stakeholder feedback, and performance reviews.

Jim used a heat map to track the strategy, which allowed him to provide the board of directors with progress updates at any time. Based on performance reviews, stakeholder feedback, and additional monitoring tools, such as dashboards displaying the progress of the work, he could accurately update the heat map. Any new risks could quickly be translated into changes that could be implemented during the next Architecture Development Method iteration (see Table 20-19).

Jim customized the Architecture Development Method for his work. Not all of the method's steps were necessary. Therefore, he only used the parts that added value to the development and implementation of the architecture they had devised. He used the TOGAF Standard as intended.